

SentinelNet

SentinelNet: AI-Powered Network Intrusion Detection System (NIDS)

A technical deep dive into optimizing model performance and deployment for real-time threat analysis in high-volume network environments.

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Core Objectives: Performance and Precision

SentinelNet's architecture prioritizes two critical factors for effective NIDS: the speed of detection and the accuracy of classification. We examine the trade-offs between various machine learning algorithms across standardized datasets.

Real-Time Latency

Minimize processing time to ensure immediate response to detected threats, crucial for mitigating fast-moving attacks like DDoS.

High Recall & Precision

Balance True Positives (high recall) and minimizing False Positives (high precision) to prevent alert fatigue and ensure genuine threats are never missed.

Resource Optimization

Efficiently manage memory, CPU, and threading to maintain performance under continuous, high-volume data streams.

Dataset Benchmark: Model Performance Comparison

Performance metrics derived from two industry-standard network intrusion datasets: NSL-KDD and CICIDS-2017. Random Forest and Gradient Boosting consistently lead in accuracy but with higher training overhead.



NSL-KDD Dataset

Focuses on traditional network features. Random Forest shows **96.2% Accuracy** but requires $\sim 3 \times$ training time.

- Logistic Regression: 89.3% Accuracy
- Decision Tree: 92.7% Accuracy



CICIDS-2017 Dataset

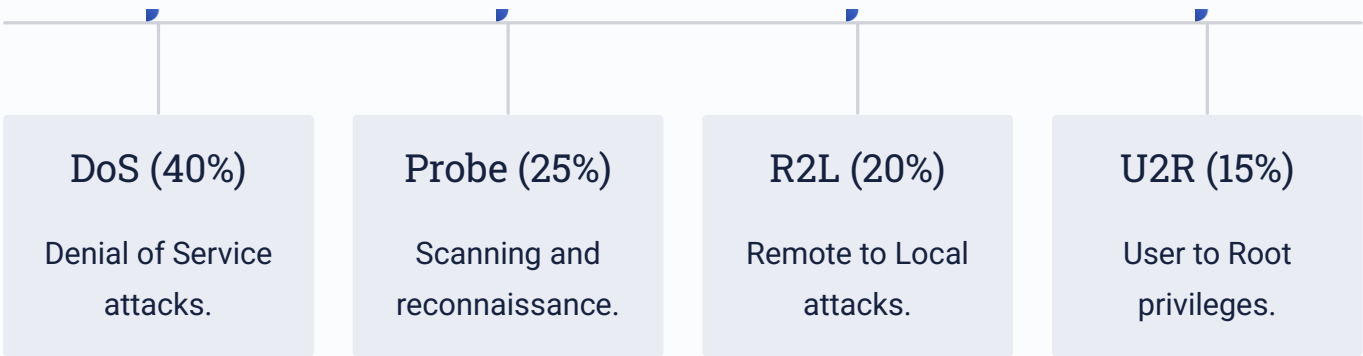
Features modern, behavior-based attributes. Random Forest achieves **99.1% Accuracy** but training extends to $\sim 3 \times$.

- Logistic Regression: 96.4% Accuracy
- Gradient Boosting: 98.9% Accuracy

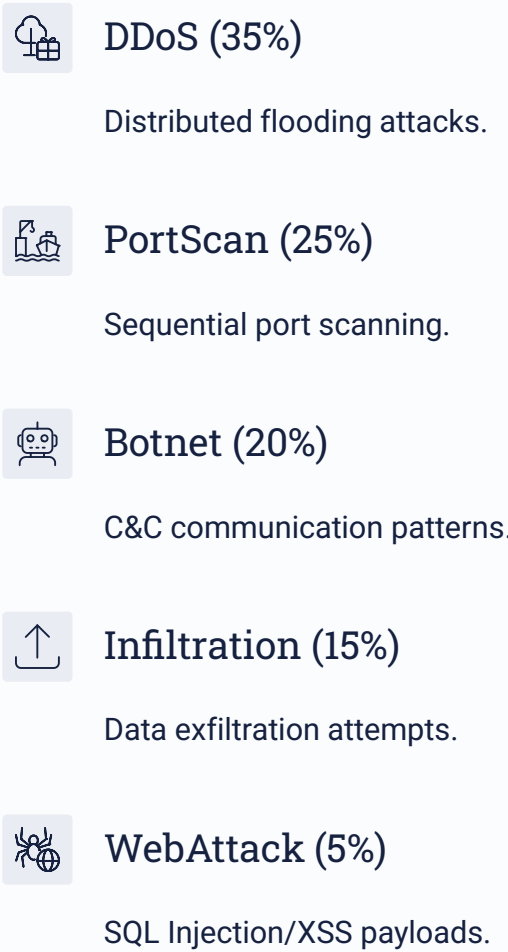
Real-World Intrusion Pattern Distribution

The type of attack varies significantly depending on the network environment and data source. Understanding the expected attack breakdown informs model selection and alert priority.

NSL-KDD Pattern Distribution



CICIDS-2017 Pattern Distribution



Optimizing Live Monitoring: The Performance Loop

To achieve low-latency detection, SentinelNet utilizes multi-threading and an optimized update interval. Simulation demonstrates handling realistic traffic volumes (3-8 packets per update cycle) with built-in metrics for false positives and negatives based on selected model recall.



Ingest & Pre-Process

Capture packets and extract features. Utilize StringIO for efficient in-memory processing.



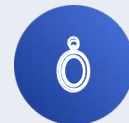
Asynchronous Prediction

Use ThreadPoolExecutor (max 2 workers) to parallelize model inference on incoming batches, maintaining responsiveness.



Alert & Log

Log detections and trigger alerts with rate-limiting ($\lambda \cdot f' \cdot f'_{L, J^{*8} \cdot l \epsilon}$) to manage system load.



Resource Management

Implement gc and history trimming (last 100 detections/50 details) to prevent memory bloat in long-running processes.

Statistical Monitoring and Visualization

Real-time statistics provide immediate operational visibility. Key metrics include total traffic volume, normal packets, detected intrusions, and the current intrusion rate. Visualizations use `plotly.graph_objects` for interactive display.

180K+

Total Packets

Cumulative number of analyzed network packets.

5.8%

Intrusion Rate

Percentage of traffic classified as malicious over time.

4.1K

Intrusions Detected

Confirmed malicious flows requiring immediate action.

2.0s

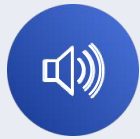
Update Interval

Frequency of data refresh and detection simulation.



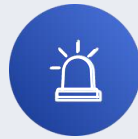
Detailed Intrusion Analysis and Severity

Beyond simple detection, SentinelNet provides granular details on source, signature, and severity. This allows security analysts to prioritize critical threats like U2R and Infiltration over standard scanning probes.



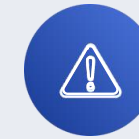
Critical Severity

Example Signatures: Distributed Flood (DDoS), Buffer Overflow (U2R), Data Exfiltration (Infiltration).



High Severity

Example Signatures: Flood (DoS), Brute Force (R2L), C&C Communication (Botnet).



Medium Severity

Example Signatures: Port Scan (Probe), Sequential Scan (PortScan).

Intrusion details capture critical metadata: Timestamp, Intrusion Type, Source IP (e.g., 192.168.100.x), Protocol (TCP/UDP), Signature, and Confidence Score.

Offline Analysis: Evaluating Model Generalization

The CSV file analysis mode allows developers and data scientists to test model performance against custom datasets or captured traffic logs. This requires robust handling of large datasets (`pd.read_csv`) and efficient sampling for performance testing.

Data Ingestion

Upload CSV; handle potential errors and display a data preview to confirm feature compatibility.

Sampling & Simulation

For large files, sample up to 1,000 records (`df.sample(n=1000)`) to simulate predictions and ensure timely metric generation.

Metric Calculation

Compute all standard evaluation metrics (Accuracy, Precision, Recall, F1) using `sklearn.metrics`.

Visualization

Generate Confusion Matrix and ROC Curve to visualize model bias and threshold effectiveness.

Deep Dive: Performance Metrics and Visualization

Robust evaluation necessitates detailed metrics. The Confusion Matrix and ROC Curve are essential for understanding how the model performs on True Positives (TP) vs. False Negatives (FN).

Evaluation Metrics

●●●●● 94.2%

Accuracy

Overall correctness across all classifications.

●●●●● 92.1%

Recall

Ability to correctly identify all relevant intrusions (minimizing FNs).

●●●●● 95.5%

Precision

Proportion of positive identifications that were actually correct (minimizing FPs).



Key Takeaways: Building a Production-Ready NIDS

The successful deployment of SentinelNet relies on strategic choices in algorithm, architecture, and monitoring feedback loops. Continuous optimization of the model is necessary as attack patterns evolve.



Prioritize Speed

Leverage asynchronous processing (`ThreadPoolExecutor`) and in-memory caching to reduce detection latency to sub-second levels.



Maximal Recall

Select models (e.g., Random Forest) and tune thresholds to minimize False Negatives, ensuring zero critical threats are missed.



Flexible Analysis

Support both live stream processing and detailed offline batch analysis for model retraining and validation against new data streams.

Thank You

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